

WASP-11b

R-0907

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_230929 · created 2026-07-29T05:26:06+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460216.8428 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.051
Transit depth (Rp/Rs)^2	3.08 +/- 1.8 [%]
Orbital Inclination (inc)	85.83 +/- 1.69
Airmass coefficient 1 (a1)	1.2 +/- 0.7
Airmass coefficient 2 (a2)	-0.2 +/- 0.57
Scatter in the residuals of the lightcurve fit is	55.62 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.02
Transit Duration (day)	0.063 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.051 vs 0.131 (deviation 0.84σ)
Duration fit vs published	0.063 d vs 0.1075 d (deviation 1.12σ)
Mid-time O–C	1.4 min
Depth SNR	3.3
Residual scatter	55.62 %

Light curve

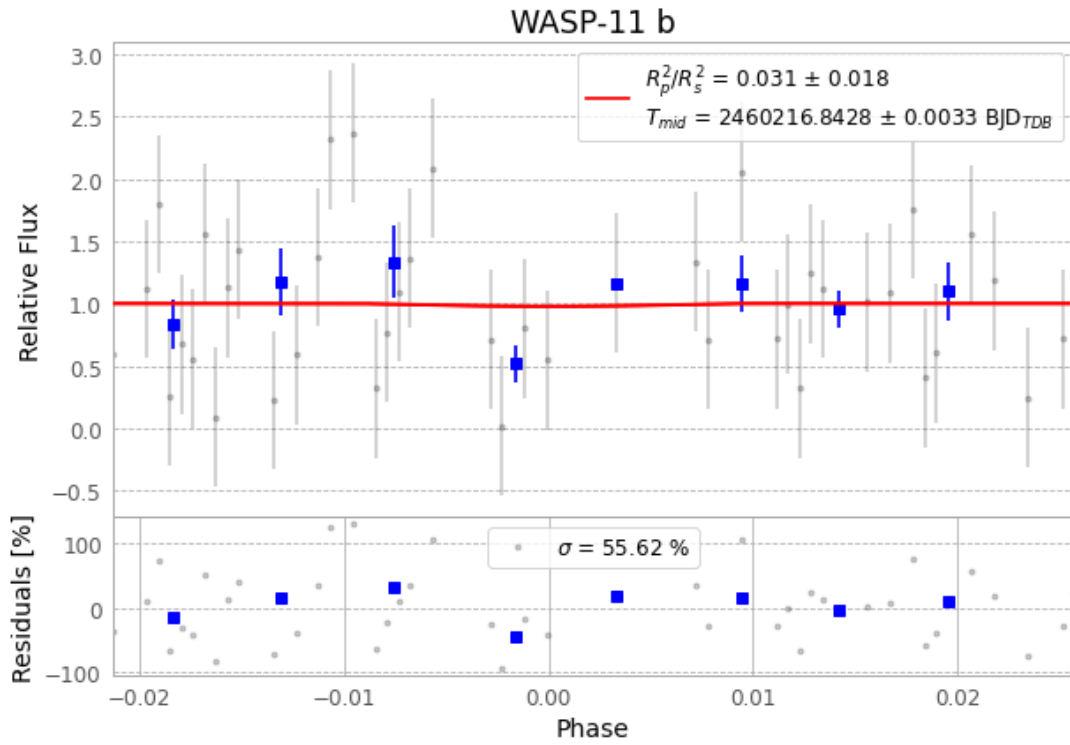


Figure 1 — FinalLightCurve_WASP-11 b_2023-09-28.png (SHA-256 09e7c9a2b58b475b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [554, 373], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 391b9a7cc811f91f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

90 FITS frames (59 MB), HATP-10230929055715.FITS → HATP-10230929102415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-230929.log (acafda2e17d5...), FinalLightCurve_WASP-11 b_2023-09-28.png (09e7c9a2b58b...), FinalLightCurve_WASP-11 b_2023-09-28.csv (75a992083c7c...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 05:26Z.